

AUTOMATIC RAILWAY GATE CONTROL SYSTEM USING IR SENSOR

FINAL PHASE 6 — FINAL SUBMISSION & PRESENTATION

Team Members: M HARINI, SM AKSHAYA, P THARANI LAKSHMI

Final Phase: Week 8 — 11 September to 14 September 2026

1. Final Phase Objective

This phase consolidates the work completed from Phase 1 to Phase 5 and prepares the team for final submission, viva, presentation, demonstration and evaluation. The final package includes technical documentation, design files, presentation materials and business & innovation content.

2. Final Project Overview

The project is a laboratory-scale Automatic Railway Gate Control System using two IR sensor pairs, an Arduino Uno controller, an SG90 servo-driven barrier, an LED and a buzzer. The base scope covers a single train approaching from one direction, gate closure, crossing occupancy and reopening after departure. Multi-track, bidirectional simultaneous traffic and multi-train scheduling are future extensions.

Phase 3 pin mapping: IR1 approach → D2; IR2 departure → D3; servo → D9; LED through 220 Ω resistor → D7; buzzer → D8; common 5 V and GND.

3. Final Project Report

- Cover page and team details
- Problem statement and abstract
- Objectives and scope
- Literature survey, existing solutions and gap analysis
- Proposed solution and requirements
- System architecture, block diagram and circuit schematic
- Component selection and BOM
- Software flowchart, algorithm and communication
- Prototype development and integration
- Testing and validation results
- Error analysis, debugging, power and reliability observations
- Limitations, future scope and references

4. User Manual

- Connect the prototype to the approved 5 V supply and check common ground.
- Power ON and allow the controller to initialize.
- Keep the sensing path clear for the normal state.
- Pass the test train/object through the approach-side IR sensor.
- Observe the programmed LED/buzzer warning and gate response.
- Continue the test through the crossing zone.
- Trigger the departure-side IR sensor after clearance.
- Observe the programmed reopening/reset sequence.

5. Installation Guide

- Mount and align both IR sensors securely.
- IR1 OUT → Arduino D2; IR2 OUT → D3.
- Servo signal → D9.
- LED → D7 through 220 Ω resistor.
- Buzzer + → D8.
- Connect the required 5 V supply and common GND.
- Check every connection before powering the prototype.
- Keep wiring protected and avoid overloaded or unsafe supplies.

6. Maintenance Guide

- Check IR sensor alignment before demonstration.
- Inspect jumper wires and breadboard connections.
- Check common ground and supply connections.
- Ensure the servo barrier moves freely.
- Keep the sensing path clean.
- Verify LED and buzzer operation.
- Use Serial Monitor logs for debugging when applicable.
- Replace damaged components only after confirming the fault.

7. Design Files Checklist

- Final Arduino source code
- GitHub repository link
- Circuit schematic
- Block diagram and flowchart
- PCB files, if applicable
- CAD/mechanical files, if applicable
- Simulation files, if applicable
- Final BOM

8. GitHub & Version Control

- Confirm the final repository is accessible.
- Maintain team-member branches.
- Team Lead reviews member pull requests.
- Faculty/Mentor reviews the Team Lead pull request.
- Keep final source code and documentation in the repository.
- Add the final repository link to the submission.

9. Final PPT Structure

- Slide 1 — Project title and team members
- Slide 2 — Problem statement
- Slide 3 — Objectives
- Slide 4 — Existing system and gap

- Slide 5 — Proposed solution
- Slide 6 — Block diagram/system architecture
- Slide 7 — Hardware and circuit
- Slide 8 — Software flowchart and algorithm
- Slide 9 — Prototype photographs
- Slide 10 — Testing and validation
- Slide 11 — Cost estimation and innovation
- Slide 12 — Limitations and future scope
- Slide 13 — Conclusion
- Slide 14 — References, GitHub and Q&A;

10. Demonstration & Explanation Video

- Keep the video within the required 3–5 minutes.
- Introduce the problem and proposed solution.
- Show the actual prototype and sensor arrangement.
- Demonstrate approach detection and gate operation.
- Demonstrate departure detection and reopening.
- Explain Arduino, IR sensors, servo, LED and buzzer.
- Use genuine prototype evidence and actual test observations.

11. Business & Innovation

Market Analysis: The present prototype is intended for educational/project demonstration. Full-scale railway deployment is outside the prototype scope and would require additional certification, weatherproofing, safety validation and railway-authority approval.

Cost Estimation: Enter the team's actual purchase prices; do not use assumed or fabricated values.

Component	Qty	Unit Cost (₹)	Total (₹)
Arduino Uno	1	[Actual]	[Actual]
IR Sensor Module	2	[Actual]	[Actual]
SG90 Servo	1	[Actual]	[Actual]
LED + 220 Ω resistor	1 set	[Actual]	[Actual]
Buzzer	1	[Actual]	[Actual]
Breadboard/Jumper wires	[Actual]	[Actual]	[Actual]
Power supply	[Actual]	[Actual]	[Actual]
TOTAL			[Actual]

Business Model: Possible target context includes educational institutions, project laboratories and engineering training demonstrations. The value proposition is a low-cost embedded prototype for demonstrating automatic train detection and gate control.

Commercialization Strategy: A field-oriented version would need rugged sensing and actuation, environmental protection, backup power, fault monitoring, safety validation, certification and approval by relevant authorities.

12. Market Pitch

“Automatic Railway Gate Control System Using IR Sensor” is a low-cost educational prototype demonstrating dual-side train detection with a microcontroller, automatic barrier control and warning indication. It provides a foundation for future engineering improvements while keeping the present prototype simple and non-networked.

13. Future Scope

- Multiple-track operation
- Bidirectional and multi-train handling
- Additional sensing such as inductive detection or RFID
- GSM/IoT remote monitoring
- Geared DC motor and limit switches for a larger barrier
- Improved redundancy and fault detection
- Backup power
- Environmental protection and field-deployment studies
- Safety validation and certification before any real-world application

14. Patent / Research Scope

Possible research directions include sensor-validation methods, improved dual-sensor fault handling, parameterized detection timing and integration of additional sensing technologies. Any patent claim should be made only after separate novelty, prior-art, inventorship and legal evaluation.

15. Viva Preparation

- Explain the problem statement and objectives.
- Explain why two IR sensors are used.
- Explain the role of Arduino Uno.
- Explain the SG90 servo's role.
- Explain LED and buzzer functions.
- Know D2, D3, D7, D8 and D9.
- Explain the complete working sequence.
- Explain debounce/validation and false-trigger issues.
- Explain prototype limitations.
- Know the actual Phase 5 test results and measurements.
- Know the actual component cost and GitHub contents.

16. Final Submission Checklist

- ■ Final Project Report
- ■ Project Documentation
- ■ User Manual
- ■ Installation Guide
- ■ Maintenance Guide
- ■ Source Code
- ■ GitHub Repository
- ■ Circuit Schematics
- ■ PCB/CAD/Simulation files where applicable
- ■ Project PPT
- ■ 3–5 minute Demonstration/Explanation Video
- ■ Working Live Prototype

- ■ Market Analysis
- ■ Actual Cost Estimation
- ■ Business Model
- ■ Commercialization Strategy
- ■ Market Pitch
- ■ Future Scope
- ■ Patent/Research Scope if applicable
- ■ Viva preparation

17. Final Conclusion

The Final Phase combines the research, concept development, system design, prototype development and validation work into the final project submission. Only genuine prototype evidence, actual test results and real development records should be submitted. The final presentation should clearly explain the system, demonstrate its operation, state its laboratory-scale limitations and present realistic future improvements.